



(12) **United States Patent**
Ball et al.

(10) **Patent No.:** **US 12,398,825 B2**
(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **TOUCHLESS UTILITY CONTROLLER**

(56) **References Cited**

(71) Applicant: **WCM Industries, Inc.**, Colorado Springs, CO (US)

U.S. PATENT DOCUMENTS

(72) Inventors: **William T. Ball**, Colorado Springs, CO (US); **Eric Pilarczyk**, Colorado Springs, WA (US); **Gabriel Young**, Colorado Springs, WA (US)

2012/0234409 A1 9/2012 Klicpera
2012/0306621 A1 12/2012 Muthu
(Continued)

FOREIGN PATENT DOCUMENTS

(73) Assignee: **WCM Industries, Inc.**, Colorado Springs, CO (US)

CA 2949757 C * 7/2020 E03B 7/071
CN 215067750 12/2021
(Continued)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

OTHER PUBLICATIONS

European Search Report for EP Application No. 23158356.8 mailed Jul. 17, 2023 (8 pages).

(21) Appl. No.: **18/742,313**

Primary Examiner — Tuyen K Vo

(22) Filed: **Jun. 13, 2024**

(65) **Prior Publication Data**

US 2024/0328535 A1 Oct. 3, 2024

Related U.S. Application Data

(63) Continuation-in-part of application No. 18/170,802, filed on Feb. 17, 2023, now Pat. No. 12,033,025.
(Continued)

(51) **Int. Cl.**
F16K 37/00 (2006.01)
G06K 7/10 (2006.01)

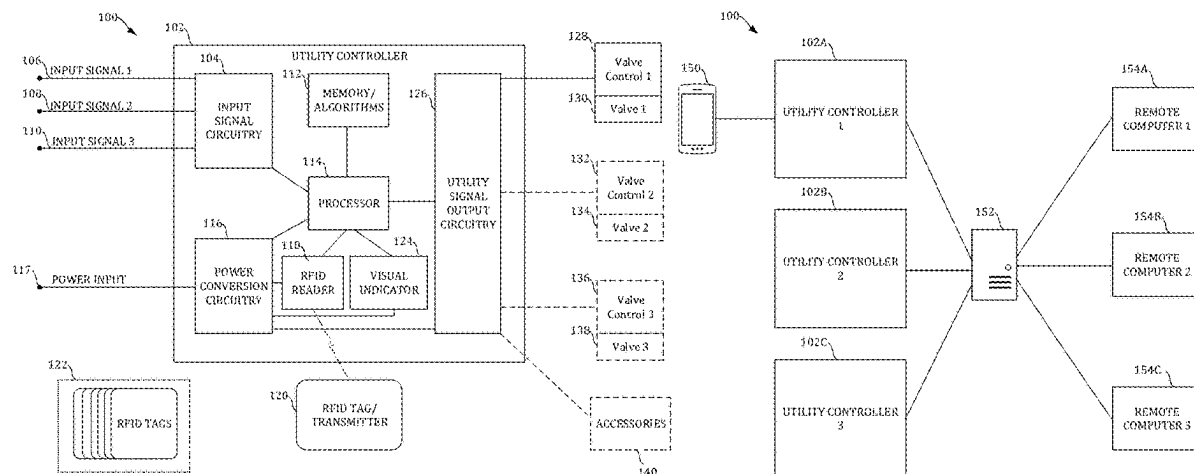
(52) **U.S. Cl.**
CPC **F16K 37/0041** (2013.01); **F16K 37/0083** (2013.01); **G06K 7/10227** (2013.01)

(58) **Field of Classification Search**
CPC F16K 37/0041; F16K 37/0083; G06K 7/10227; G06K 17/0022; G06Q 10/063; G06Q 50/06; G06Q 50/163; H04B 5/77
See application file for complete search history.

(57) **ABSTRACT**

A touchless utility controller is provided. In examples, the utility controller comprises a housing encompassing circuitry configured to receive a signal, and based on the received signal, provide an activation signal to a valve control device for a valve. The housing may comprise a face having an illuminated portion to allow for a visual indicator to illuminate the illuminated portion, wherein the face does not include any physically activated interface elements. In addition, the housing may include at least one of an outer side perimeter connected to the face or a back side, wherein at least one of the outer side perimeter or the back side defines at least one through hole for receiving a input power wiring and output transmission wires for connecting to the valve control device for the valve controlling one of gas or water.

8 Claims, 8 Drawing Sheets



Related U.S. Application Data

- (60) Provisional application No. 63/325,363, filed on Mar. 30, 2022, provisional application No. 63/314,150, filed on Feb. 25, 2022.

References Cited

(56)

U.S. PATENT DOCUMENTS

2014/0313016 A1 10/2014 Nizam
 2015/0216369 A1 * 8/2015 Hamilton G08B 21/245
 222/23
 2015/0315008 A1 11/2015 Locke et al.
 2015/0370225 A1 12/2015 McCune
 2018/0119976 A1 * 5/2018 Kadah H03K 17/74
 2019/0151879 A1 5/2019 Cortland et al.
 2019/0280507 A1 9/2019 Dallas
 2021/0347628 A1 * 11/2021 Bendig B67D 1/0888
 2021/0401670 A1 * 12/2021 Abal A61M 5/172
 2022/0260084 A1 8/2022 Mariano
 2023/0032735 A1 * 2/2023 Friend F24F 11/63

FOREIGN PATENT DOCUMENTS

EP 2615432 A1 * 7/2013 G01M 3/3227
 EP 2700854 A2 2/2014
 JP 2018526953 9/2018
 KR 20100059149 6/2010
 KR 20170091421 8/2017
 WO WO-2023043403 A1 * 3/2023 E03C 1/023

* cited by examiner

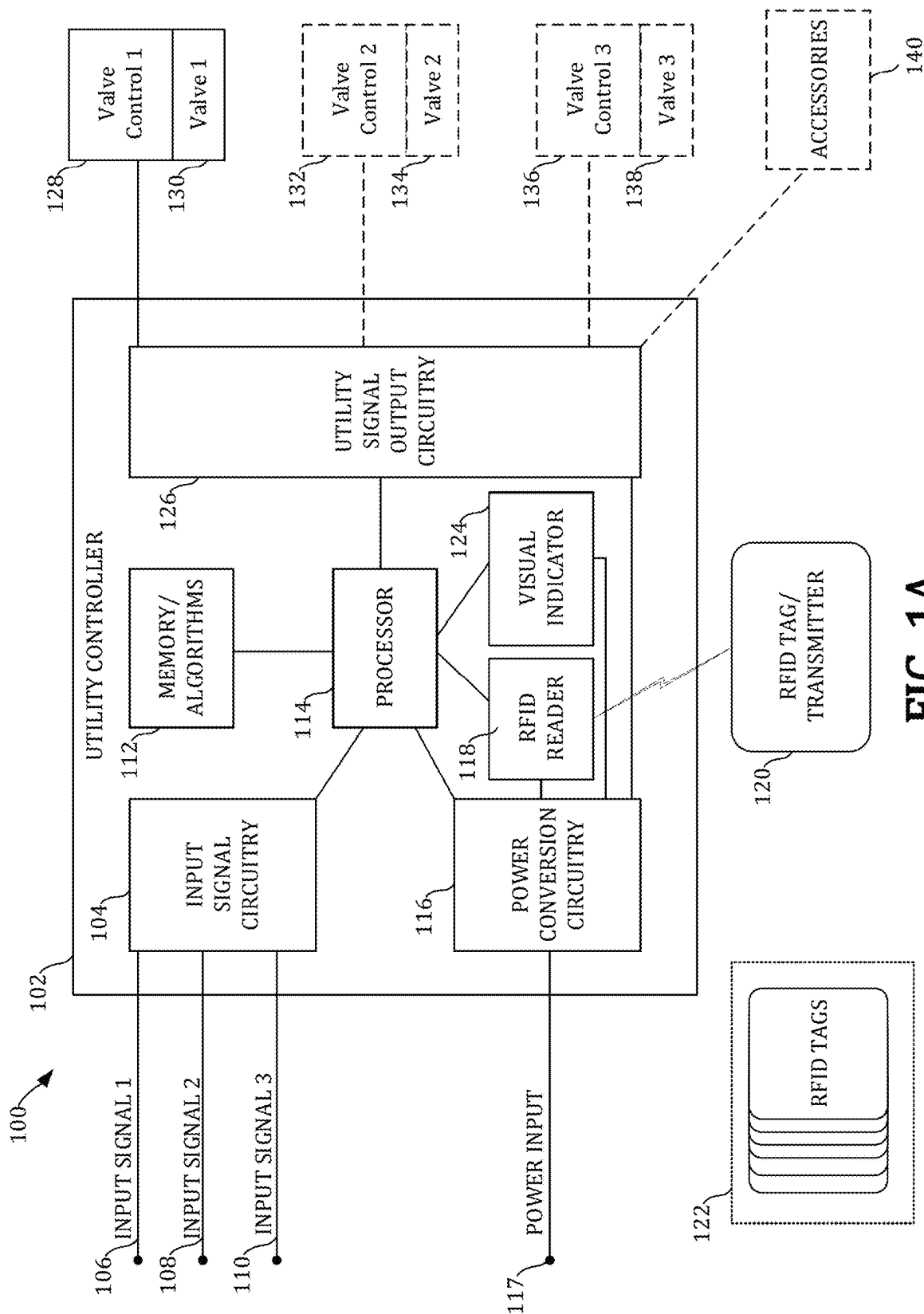
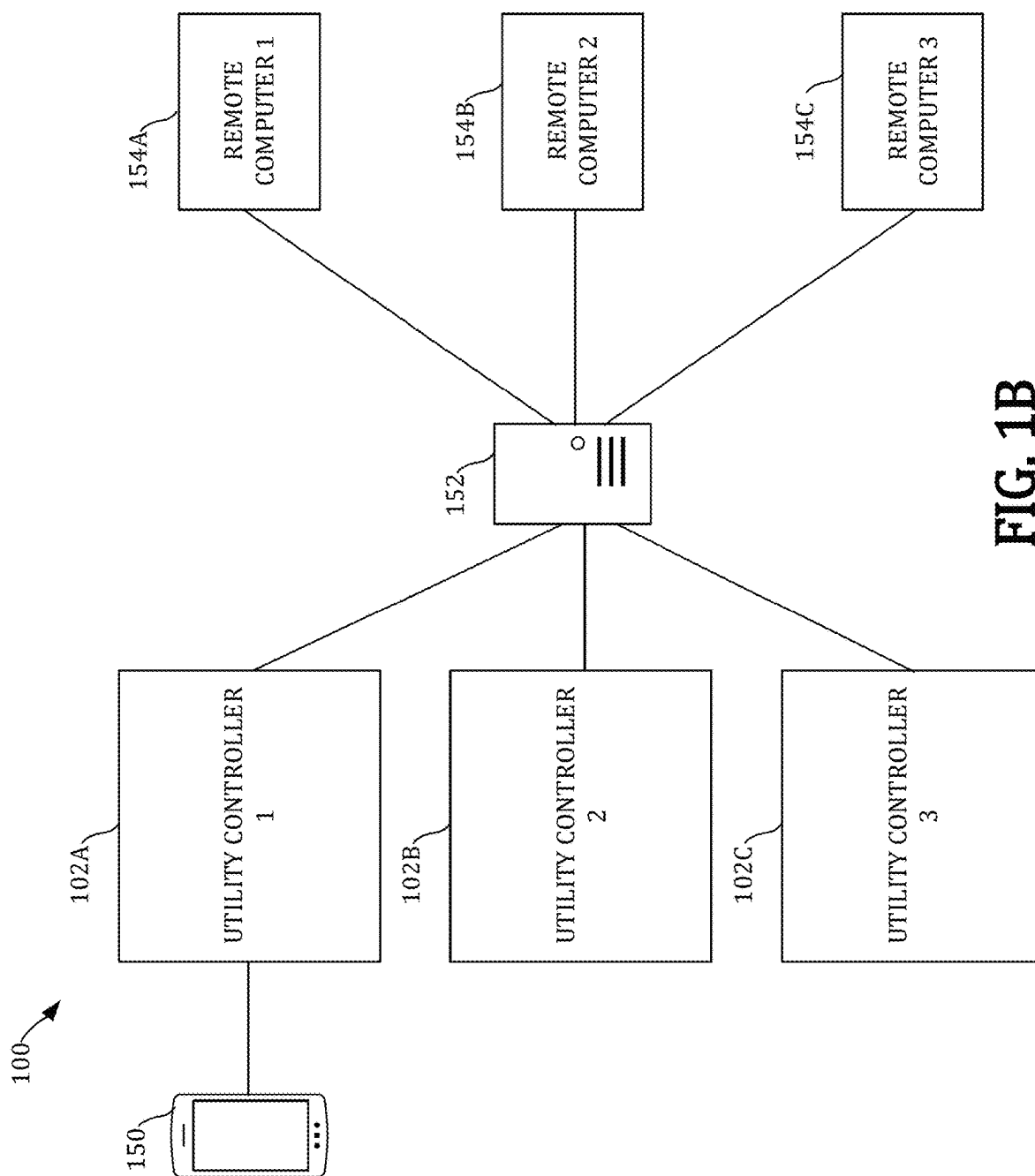
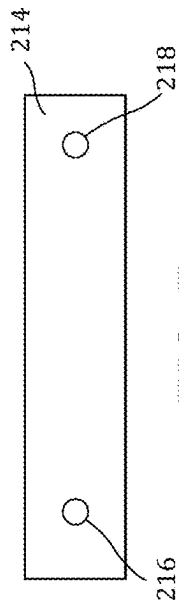
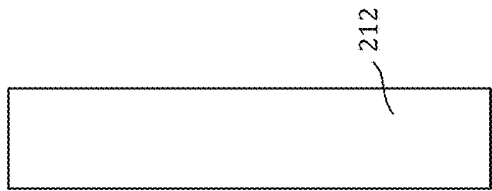
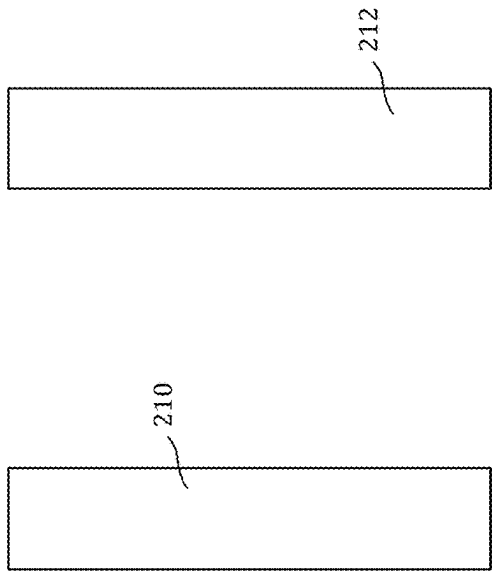
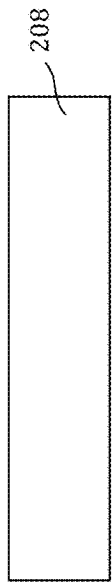
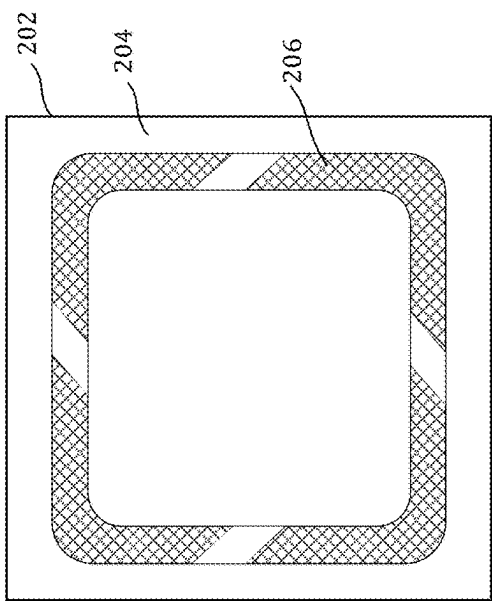
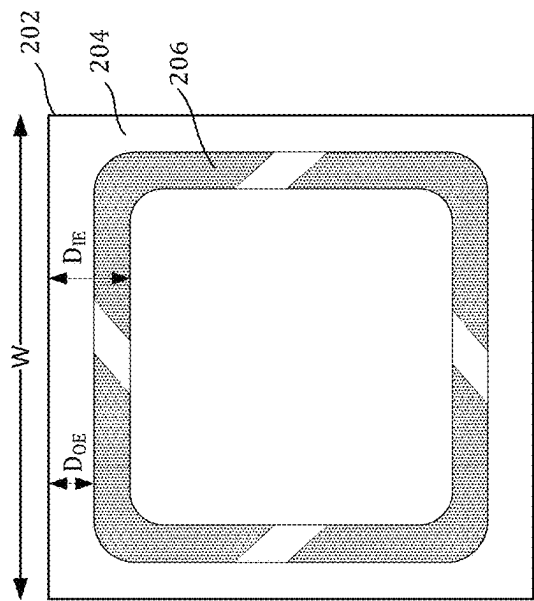


FIG. 1A





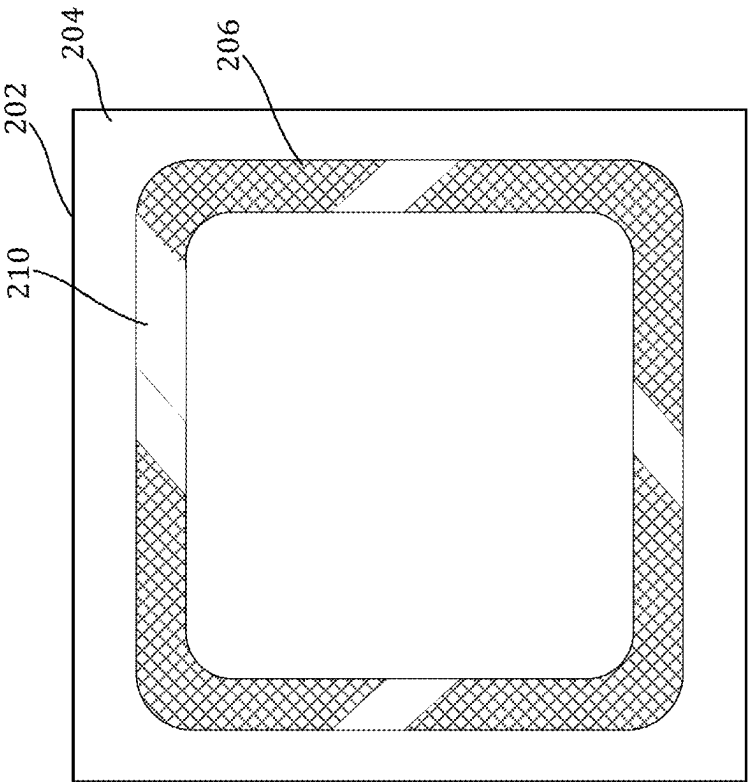


FIG. 9

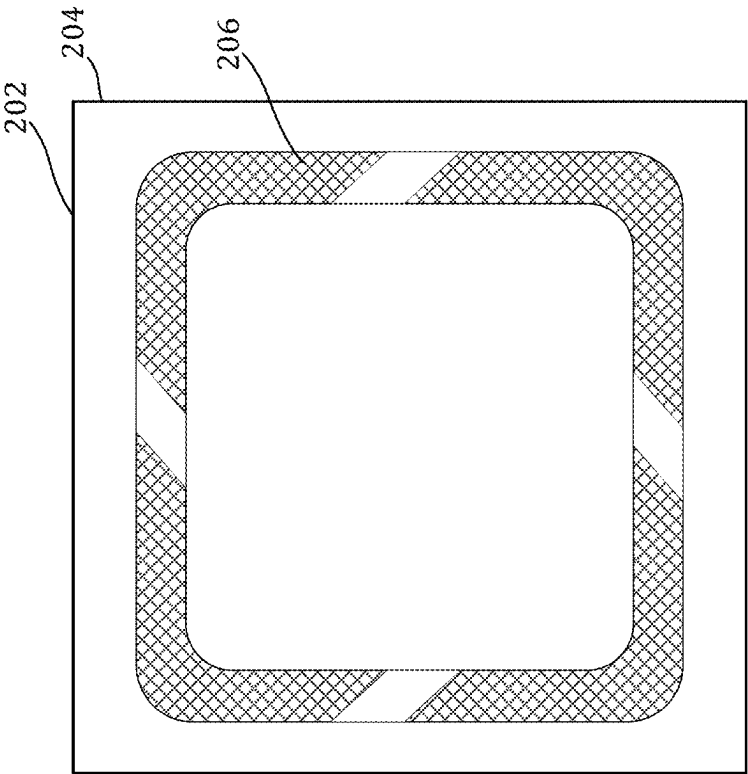


FIG. 8

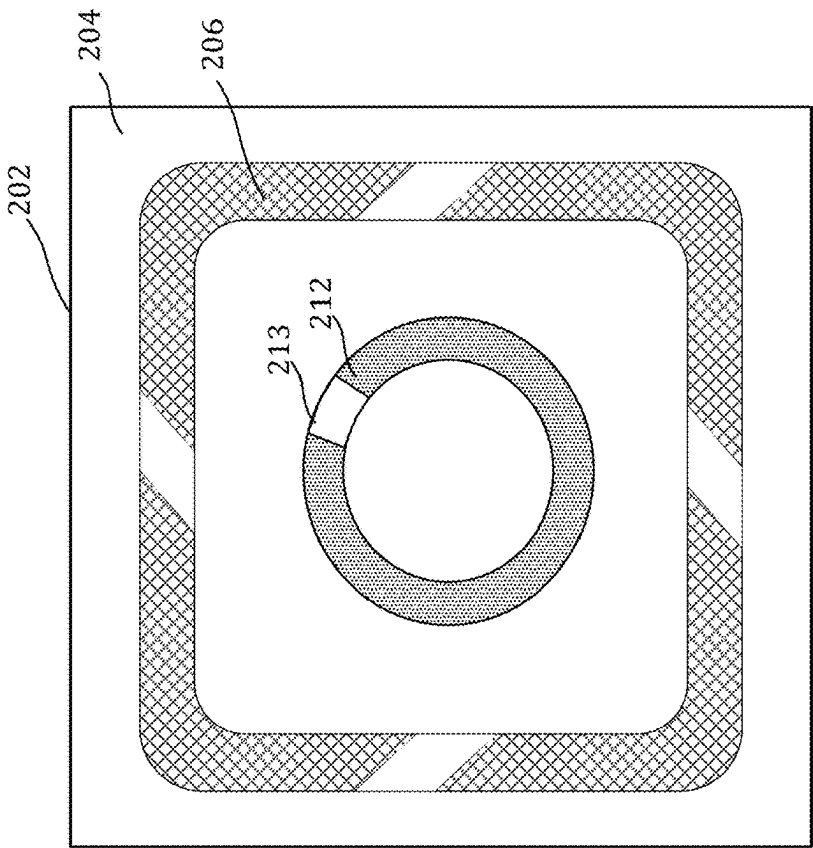


FIG. 11

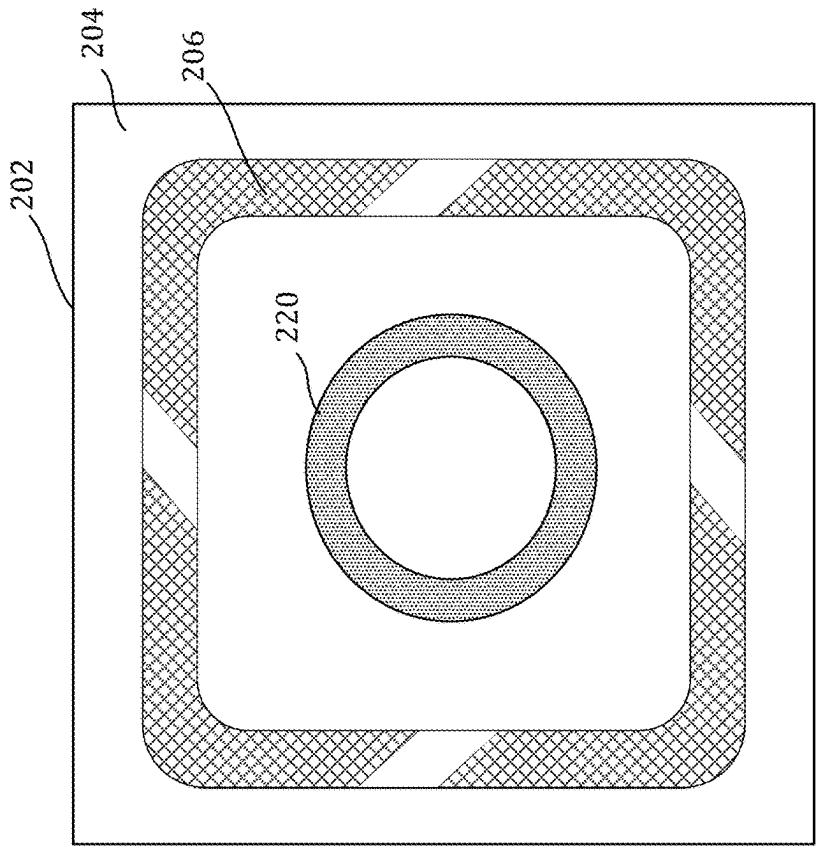


FIG. 10

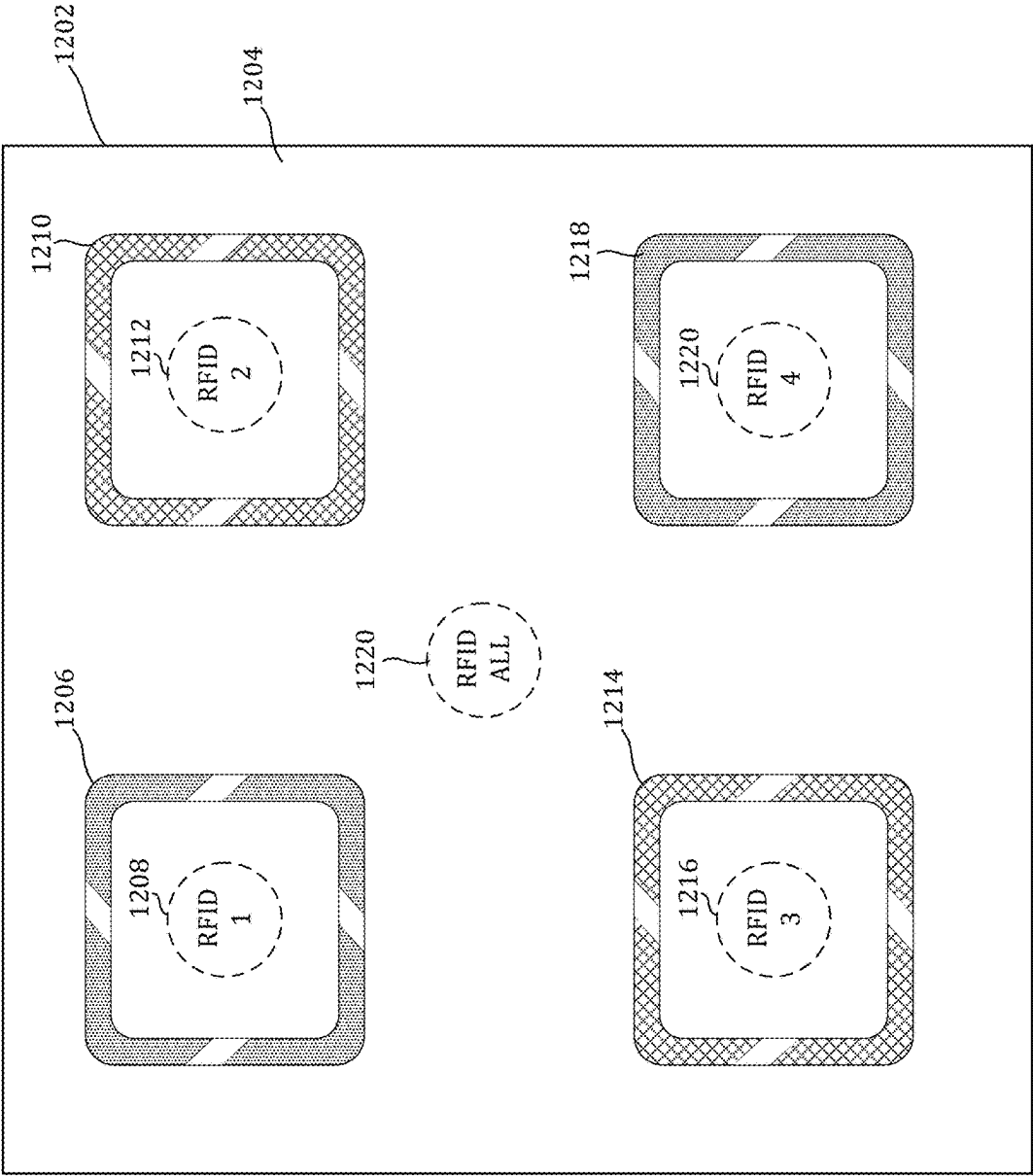


FIG. 12

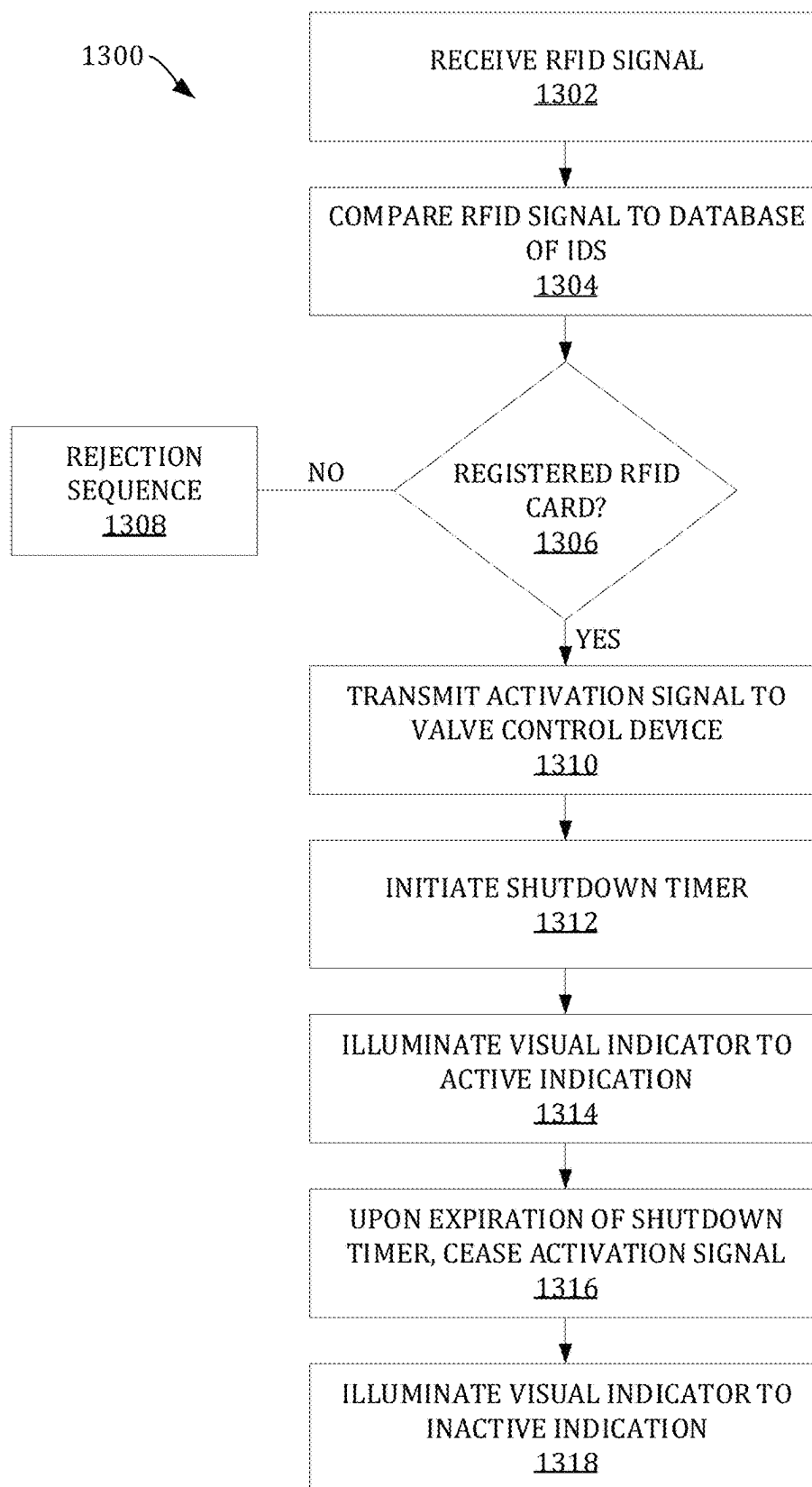


FIG. 13

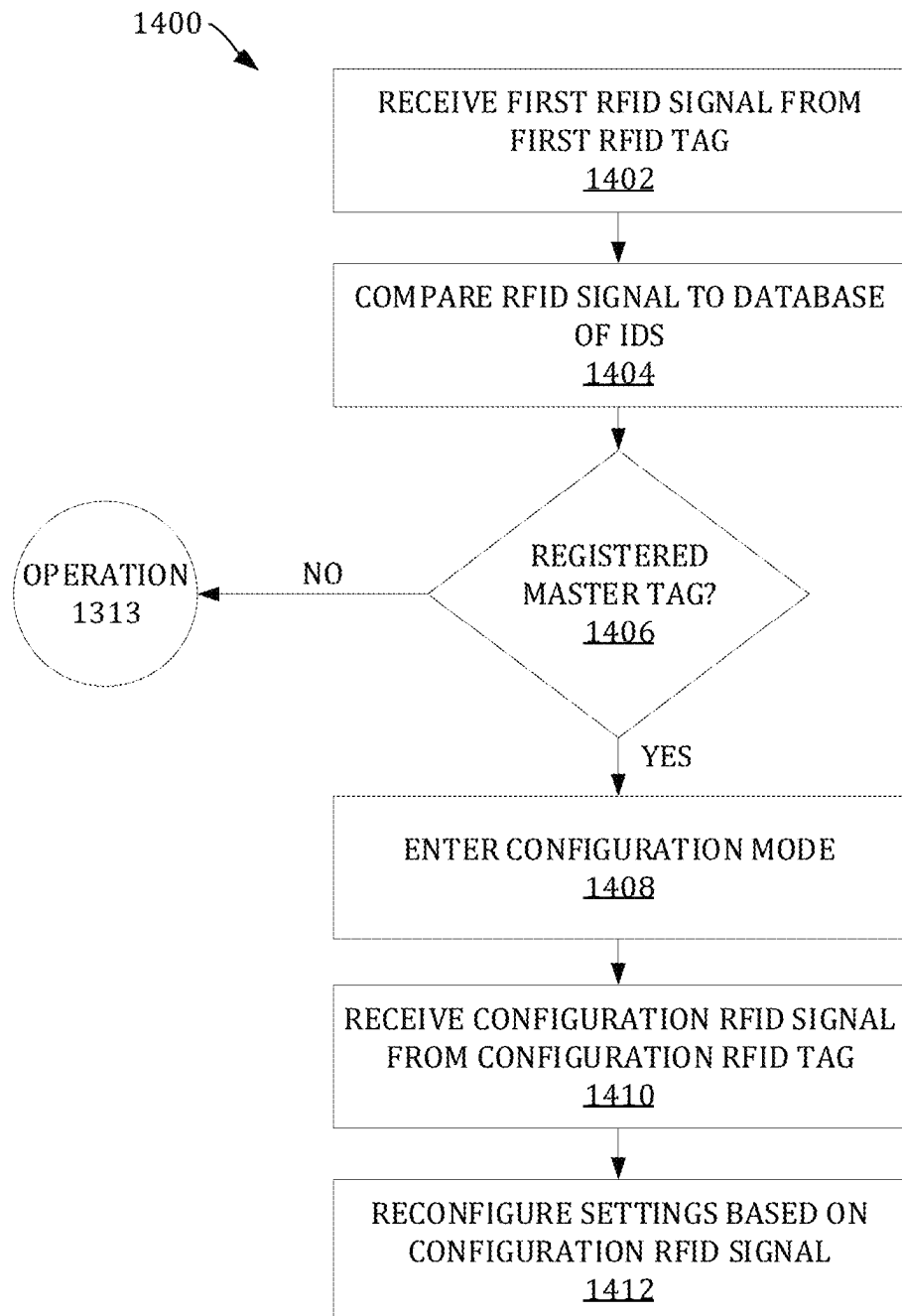


FIG. 14

TOUCHLESS UTILITY CONTROLLER**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation-in-part of U.S. patent application Ser. No. 18/170,802 filed on Feb. 17, 2023, which claims the benefit of U.S. Provisional Application No. 63/314,150 filed 25 Feb. 2022, U.S. Provisional Application No. 63/325,363 filed 30 Mar. 2022, all of which applications are incorporated herein by reference in their entireties. To the extent appropriate a claim of priority is made to each of the above-referenced applications.

BACKGROUND

Utilities such as gas, water, and electricity may need to be turned on or off in many different situations. For instance, in laboratories, restaurants, schools and other environments, providing natural gas to the corresponding equipment during times when the gas is not needed (e.g., outside of business hours) may not be preferable. Water and/or electricity may be similarly be unnecessary during such times.

It is with respect to these and other general considerations that the aspects disclosed herein have been made. Also, although relatively specific problems may be discussed, it should be understood that the examples should not be limited to solving the specific problems identified in the background or elsewhere in this disclosure.

SUMMARY

Examples of the present disclosure describe systems and methods for controlling utilities through a touchless utility controller.

In an example, the present disclosure describes a system for controlling a utility. In examples, the system comprises an RFID reader; a visual indicator; a processor; and memory, storing list of registered unique identifiers and storing instructions that, when executed the processor, cause the system to perform operations. In examples, the operations include: receiving, by the RFID reader, an RFID signal from an RFID tag, the RFID signal including a unique identifier; comparing the unique identifier to the list of registered unique identifiers to determine that the RFID tag is a registered RFID tag; in response to determining the RFID tag is a registered RFID tag: transmitting an activation signal to a valve control device to cause a valve controlling a utility to open; illuminating the visual indicator to a first color; and initiating a shutdown timer; and upon expiration of the shutdown timer: ceasing the transmission of the activation signal; and illuminating the visual indicator to a second color.

In another example, the present disclosure describes a touchless utility controller. In examples, the touchless utility controller comprises a housing encompassing: circuitry configured to receive a signal, and based on the received signal, provide an activation signal to a valve control device for a valve. In examples, the housing also comprises: a face having an illuminated portion to allow for a visual indicator to illuminate the illuminated portion, wherein the face does not include any physically activated interface elements; and at least one of an outer side perimeter connected to the face or a back side. In examples, at least one of the outer side perimeter or the back side defines at least one through hole for receiving an input power wiring and output transmission

wires for connecting to the valve control device for the valve controlling one of gas or water.

In another example, the present disclosure describes a method for controlling a utility, comprising: receiving, by an RFID reader of a utility controller, a first RFID signal from a first RFID tag; and comparing, by a processor of the utility controller, the first RFID signal to a list of registered unique identifiers to determine that the first RFID tag is a registered RFID tag. In response to determining the first RFID tag is a registered RFID tag, the method may further comprise: transmitting an activation signal to a valve control device to cause a valve controlling a utility to open; illuminating a visual indicator to a first color; and initiating a shutdown timer. In examples, the method may further comprise receiving, by the RFID reader, a second RFID signal from a second RFID tag; comparing, by the processor of the utility controller, the second RFID signal to the list of registered unique identifiers to determine that the second RFID tag is a registered RFID tag; and in response to determining the second RFID tag is a registered RFID tag: ceasing the transmission of the activation signal; and illuminating the visual indicator to a second color.

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Additional aspects, features, and/or advantages of examples will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

Non-limiting and non-exhaustive examples are described with reference to the following figures.

FIG. 1A depicts an example system for controlling at least one utility with a touchless utility controller.

FIG. 1B depicts additional components of an example system for interacting with a touchless utility controller using one or more mobile device, server, and/or remote computing device.

FIG. 2 depicts a front view of a touchless utility controller with a visual indicator in an inactive state.

FIG. 3 depicts a front view of the touchless utility controller with a visual indicator in an active state.

FIG. 4 depicts a top view of the touchless utility controller.

FIG. 5 depicts a left side view of the touchless utility controller.

FIG. 6 depicts a right side view of the touchless utility controller.

FIG. 7 depicts a bottom view of the touchless utility controller.

FIG. 8 depicts a front view of the touchless utility controller with a visual indicator that changes with a shutdown timer.

FIG. 9 depicts a front view of the touchless utility controller with the visual indicator that changes with the shutdown timer after a partial duration of the shutdown timer has passed.

FIG. 10 depicts a front view of the touchless utility controller with a visual indicator and a shutdown timer indicator.

3

FIG. 11 depicts a front view of the touchless utility controller with the visual indicator and the shutdown timer indicator after a duration of the shutdown timer has passed.

FIG. 12 depicts a front view of a touchless utility controller with multiple RFID readers and visual indicators for different utilities.

FIG. 13 depicts an example method for controlling a utility with a touchless utility controller.

FIG. 14 depicts an example method for configuring a touchless utility controller.

DETAILED DESCRIPTION

As briefly discussed above, control of one or more utilities, such as gas, water, and electricity, may be beneficial in many different settings. For instance, in a restaurant or kitchen setting, it may be desirable to remove natural gas from the cooking appliances outside of business hours. Similarly, for a laboratory or a classroom, having running water and/or gas delivered to different stations (e.g., sinks, burners, etc.) may not be necessary outside of school or lab hours. In addition, having such utilities remain on and providing the gas and/or water to the appliances and/or stations may increase the risk of potential damage caused by flooding, fires, or other potential disasters.

One option for controlling utilities requires a manual activation of a valve (e.g., manually rotating a valve between an open and closed position) at a main source of the gas or water. While such an option may be simple and reliable, such a manual option is often not possible as the valves are not readily accessible to most operators, such as lab technicians, teachers, chefs, etc. Another option for controlling utilities is to provide a complex system that allows for significant physically selected options (e.g., buttons, touchscreens, dual in-line package (DIP) switches, physical keys) and configurations by an operator to control the activation of utilities remotely. However, such complexity significantly increases costs and manufacturing difficulties. Further, such complexity is only useful where the operator is well-trained on how to operate the device. For example, systems that require physical keys and selections of multiple buttons may potentially cause uncertainty and discomfort for the operator—especially where the control of flammable substances like natural gas is involved. As a result, such complex controllers often go unused and/or the corresponding utilities are not controlled in the intended manner.

The present technology, among other things, may address the above issues by providing an intuitive and reliable utility controller that may utilize no buttons, keys, touchscreens, touch inputs, DIP switches or other physically activated interface elements on the face or front of the utility controller. Instead of using such physically activated interface elements, the present utility controller may implement short-range wireless communication, such as radio frequency identification (RFID), to activate and deactivate the corresponding utilities. For instance, when an operator holds a pre-registered RFID tag near the utility controller, the utility controller either activates or deactivates a utility. As an example, a chef may enter a kitchen to begin a shift and hold the registered RFID tag near the utility controller to activate the natural gas to the kitchen. At the end of the day, the chef may again hold the registered RFID tag near the utility controller to deactivate the natural gas. The utility controller may also include a shutdown timer that is initiated when the utility is activated. Upon expiration of the shutdown timer, the utility controller automatically deactivates the utility. Accordingly, in the example of the chef above, even if the

4

chef forgot to deactivate the natural gas when leaving for the day, the utility controller will automatically deactivate the natural gas upon expiration of the shutdown timer.

Such touchless control of the utility controller also provides for additional mechanical advantages in both physical space savings for the utility controller and the ability to better seal and water/weather proof the utility controller. For instance, by not having any protruding physical switches through the housing, the likelihood of water or other environmental elements from entering the housing is significantly reduced. As a result, the utility controller may also be used in outdoor environments as well as indoor environments. Moreover, without any keys, physical switches, or buttons on the utility controller, the utility controller may be manufactured without any moving parts. With no moving parts or direct manual interaction with the controller by operators, the utility controller is less likely to suffer from wear-and-tear, which further improves the longevity of the utility controller.

The utility controller may also include a visual indicator that clearly indicates the on or off state of the corresponding utility, even from a distance. For example, the face of the utility controller may include a large visual indicator that changes color based on the current state of the utility controller. By changing color rather than simply turning off the visual indicator, confusion is removed as to whether power has been lost, the controller is not functioning, and/or the state of the utility has changed. The combination of the streamlined wireless interface with the clear visual indicator provides for an improved utility controller that is intuitive, reliable, and instills confidence in the operator to correctly use the utility controller.

FIG. 1A depicts an example system 100 for controlling at least one utility with a touchless utility controller 102. The utility controller 102 includes a plurality of components that allows for the selective control of one or more electronic devices or utilities, such as natural gas, water, or electricity.

The utility controller 102 may include input signal circuitry 104 for receiving input signals into the utility controller 102. The input signal circuitry 104 may include physical components and well as signal conditioning and/or modifying components that modify the incoming input signals to a format that may be processed or used by the utility controller 102.

The input signals may include a variety of signals, such as a first input signal 106, a second input signal 108, and a third input signal 110. The input signals may include a panic button activation signal, which may be a signal generated from the manual activation of a panic button (not shown). The panic button may be located at a position remote from the utility controller 102, such as on another side of a room or building. The input signals may also or alternatively include a building management system signal that may indicate hours of operation of the building or other similar signal that may be used by the utility controller 102 to activate or deactivate one or more of the utilities based on the building policies or controls. The input signals may also or alternatively include an emergency service signal that may be generated from activation of a fire alarm, carbon monoxide monitor, or other similar devices.

The utility controller 102 also includes a processor 114 that processes data and controls the operation of the utility controller 102. For instance, the processor 114 receives the input signals from the input signal circuitry 104 and processes those signals to determine actions and outputs of the

input signal circuitry **104**. The processor **114** also receives signals from other sources, as depicted in FIG. 1A and discussed further below.

A memory **112** is also included in the utility controller **102**. The memory **112** may be a non-transitory, physical storage medium. The memory **112** stores instructions corresponding to algorithms that, when executed by the processor **114**, cause the utility controller **102** to perform the operations discussed herein among other potential operations. For instance, the memory **112** may store instructions that, when executed by the processor **114**, cause the utility controller **102** to perform the operations of the methods **1300** and **1400** discussed below with reference to FIGS. **13-14**. In some examples, the memory **112** and the processor **114** may be packaged in the same chip or component, such as in a microcontroller or other similar device.

The utility controller **102** may also include utility signal output circuitry **126**. The output circuitry **126** may include physical components for mechanical connection of wiring between the processor **114** and outputs to valves and/or other accessories **140**. The output circuitry **126** may also include signal modifying and/or conditioning components that modify a received signal. When the output circuitry **126** receives an activation signal for a particular utility, the output circuitry **126** generates a new signal, modifies the activations signal, and/or passes the activation signal to one or more valves corresponding to the utility to be activated. The output signals from the output circuitry **126** may be high voltage (e.g., 60 volts or higher) or low voltage (e.g., 24 volts or less) depending on the type of valve control device that is used to control the various valves.

In some examples, the utility controller **102** may control a single valve of the system **100**, such as a first valve **130**. The first valve **130** may be a valve that controls the flow of gas or water. The position of the first valve **130** is controlled by a first valve control device **128**, which may be a solenoid, a motor to rotate a ball valve, or other type of valve actuator. When the first valve control device **128** receives the activation signal from the output circuitry **126**, the first valve control device **128** causes the first valve **130** to open, which results in the corresponding fluid (e.g., water or gas) to flow through the first valve **130**. When the activation signal ceases, the first valve control device **128** causes the first valve **130** to close, which prevents the flow of the corresponding fluid.

In the example of a solenoid, the solenoid may be a normally closed solenoid that is held open while the activation signal is present. Thus, when the activation signal ceases or is removed, the solenoid returns to its normally closed state. In the example of a motor to control a ball valve, the activation signal may cause the motor to rotate the ball of the ball valve to an open position. While the activation signal is present, the power from the signal charges a capacitor in the first valve control device **128**. When the activation signal ceases or is removed, the capacitor discharges and causes the motor to rotate the ball of the ball valve to a closed position. In examples, the output circuitry **126** may be configured to output the activation signal only on demand (thus resulting in the first valve **130** being normally closed and the utility being normally off). In other examples, the output circuitry **126** may be configured to cause the activation signal to be normally present (e.g., on startup of controller **102**), and the solenoid may cause the first valve **130** to be held open (and the utility on) unless and until the activation signal is removed in response to receiving a signal to turn the utility off (e.g., from the RFID reader

118 reading a tag **120**, receiving an emergency system input, receiving input from a panic button, etc.).

In other examples, the first valve **130** may be biased to be normally open, and an activation signal from circuitry **126** causes the first valve **130** to close. In the solenoid example, the solenoid may be a normally open solenoid that is only held closed while the activation signal is present. Thus, when the activation signal ceases or is removed, the solenoid returns to its normally open state. In the example of a motor to control a ball valve, the activation signal may cause the motor to rotate the ball of the ball valve to a closed position. While the activation signal is present, the power from the signal charges a capacitor in the first valve control device **128**. When the activation signal ceases or is removed, the capacitor discharges and causes the motor to rotate the ball of the ball valve to an open position.

Additional valves and valve control devices may also be included in the system **100** and controlled by the utility controller **102**. For instance, a second valve control device **132** and a second valve **134** may be included. A third valve control device **136** and a third valve **138** may also be included. The second valve control device **132**, second valve **134**, third valve control device **136**, and the third valve **138** may be the same or different types of devices or valves as the first valve control device **128** and the first valve **130**.

The additional valves may be for controlling the same utility or a different utility. In an example, the first valve **130**, second valve **134**, and the third valve **138** may all control natural gas. For instance, each valve may control natural gas to a different station in a laboratory or to a different major appliance in a kitchen. Accordingly, a common activation signal may be sent to all three of the valves. Alternatively, the valves may be individually controlled with different activation signals sent and controlled for each of the valves.

In other examples, the different valves may control different utilities. For example, the first valve **130** may control the flow of gas and the second valve **134** may control the flow of water. In such an example, different activation signals may be generated for control of the first valve **130** and control of the second valve **134**.

One or more accessories **140**, such as a fan or other electronic accessories, may also be controlled by the utility controller **102**. In some examples, a fan of a vent hood may be desired to be activated in all cases where the natural gas is activated. In such examples and where the first valve **130** controls the flow of natural gas, activation signals may be concurrently sent to the first valve control device **128** and the fan **140** when a natural gas utility is selectively activated by an operator. The electronic accessories **140** may be controlled by providing and removing power from the accessories **140**. In other examples, an activation signal may be sent to an electrically controlled device, such as a relay or a switch, that causes the accessory **140** to turn on or off and/or allow electrical power to be provided to the accessory **140**. In such examples, the electrically controlled device for controlling the accessory **140** provides a similar function as the valve control devices discussed above. Accordingly, in some examples, the valve control devices may also be considered electrically controlled devices.

The utility controller **102** also includes an RFID reader **118**. The RFID reader **118** may include an antenna and a transceiver. The RFID reader **118** may generate an RFID-activate signal that is transmitted wirelessly and activates an RFID transponder of an RFID tag **120** of a plurality of RFID tags **122**. The transponder of RFID tag **120** then returns an RFID signal with RFID data of the RFID tag **120**, which may include a unique identifier for the RFID tag **120**. The

data received by the RFID reader **118** from the RFID tag **120** may then be processed by the processor **114** to determine actions to be taken by the utility controller **102**.

In examples where the RFID data in the received RFID signal includes a unique identifier for the RFID tag **120**, the unique identifier may be compared to set of registered identifiers stored in the memory **112**. The registered identifiers may correspond to RFID tags that have been previously authorized to activate or deactivate a particular utility. In some examples, some RFID tags in the set **122** may be registered to control a first utility or valve and other RFID tags in the set **122** may be registered to control a second utility or valve. The registered identifiers may be loaded into the memory **112** when the utility controller **102** is manufactured, shipped, and/or installed. For instance, the utility controller **102** may be delivered with the set of RFID tags **122** that include RFID tags with the registered unique identifiers. Accordingly, those registered RFID tags may be given to the operators of the utility controller **102**, and access to the control of the corresponding utility may be controlled by limited those persons that receive a registered RFID tag. In some examples, additional RFID tags may be added to list the registered RFID tags through the use of a master RFID tags or other techniques discussed herein.

When the RFID reader **118** reads a registered RFID tag **120**, the processor **114** causes an activation signal to be generated and transmitted to one or more of the valve control devices **128**, **132**, **136** to activate a corresponding utility by opening one of the corresponding valves **128**, **132**, **136**. As an example, when a registered RFID tag **120** is detected, the utility controller **102** sends an activation signal to the first valve control device **128** that causes the first valve **130** to open. While the first valve **130** is open, the corresponding utility is in an active or “on” state. The processor **114** may also initiate a shutdown timer having a set duration. The set duration may be based on the particular RFID tag **120** that is detected or a setting of the utility controller **102**.

The valve remains open and the utility remains in the active state until one or more deactivation conditions occur. The deactivation conditions may include receiving a subsequent detection of the registered RFID tag **120** (or another registered RFID tag), expiration of a shutdown timer, receiving a signal from an emergency services input signal, receiving a signal from a building management services input signal, and/or receiving a signal from a panic button input signal. In examples, when a deactivation condition occurs, the generation and/or transmission of the activation signal to the first valve control device **128** is ceased, which causes the first valve **130** to close and the corresponding utility to enter an inactive or “off” state. In examples, the first valve **130** may be closed due to an emergency situation (e.g., receiving an emergency shutdown signal in the form of, e.g., an emergency services input signal, a signal from a panic button, etc.). In some examples, the utility controller **102** will not allow the valve **130** to be reopened until an emergency override signal is received from an authorized operator. For example, the utility controller **102** may receive an emergency override signal in the form of a second emergency services input signal, or the RFID reader **118** may receive an emergency override signal in the form of reading an RFID tag **120** registered to an authorized user having elevated privileges, or an RFID tag **120** that is specifically designated as an emergency override tag, etc.

Beyond activation and deactivation of the utility, the utility controller **102** also uses the set of RFID tags **122** to modify settings of the utility controller **102**. For example, a master RFID tag may be provided in the set of RFID tags

122. The master RFID tag transmits a unique identifier that corresponds to a master functionality in the list of registered identifiers in the memory **112**. When the master RFID tag is scanned (e.g., the RFID reader **118** reads/detects the master RFID tag), the utility controller **102** then enters a configuration mode and waits for an additional RFID tag to be scanned. For example, an unregistered RFID tag or a configuration RFID tag may be scanned while the utility controller **102** is in the configuration mode.

When an unregistered RFID tag is scanned during the configuration mode, that unregistered RFID tag is added as a registered RFID tag. For instance, the identifier corresponding to the unregistered RFID tag may be added to the list of registered identifiers.

When the configuration RFID tag is scanned during the configuration mode, one or more settings of the utility controller **102** may be changed. The configuration RFID tag, which may be in the set of RFID tags **122**, may have another unique identifier and/or store configuration data that is transmitted to the RFID reader in the configuration RFID signal. For example, the processor **114** may identify the configuration RFID tag, and based on the identification of the configuration RFID tag, perform a local lookup to determine a corresponding settings change that should be made. In other examples, the configuration RFID signal may include data directly indicating the settings change that should be made. For example, the settings change may be a change to a shutdown timer duration, and the configuration data may indicate a particular duration for the shutdown timer.

Some changes to the configuration or settings of the utility controller **102** may also be achieved through the master RFID card alone based on the duration that the master RFID card is scanned. For example, if the master RFID card is scanned (e.g., held near the utility controller **102**) for a duration exceeding a reset threshold (e.g., 30 seconds, 1 minute), the utility controller may perform a factory reset of the utility controller **102**. The factory reset may also clear from memory **112** the registered identifiers for RFID cards (other than the master RFID cards or some subset of RFID cards).

In examples, as used herein, RFID refers to short-range radio frequency communication. In examples, short-range radio frequency communication refers to radio communication with less than one foot, less than three feet, less than ten feet, or less than twenty feet of effective range between an RFID tag or emitter and an RFID reader. In examples, as used herein, RFID also encompasses near-field communications (NFC) protocols, or other similar protocols for short-range radio frequency communication. The RFID tag **120** may be in the form of a card, a fob, or other type of physical housing for the RFID tag **120**. In some examples, the RFID tag **120** may also be digital tag, such as provided by a mobile device (e.g., smartphone). For instance, the mobile device may be configured to provide the RFID data to the RFID reader **118** when the mobile device is brought into proximity with the utility controller **102** (which may be triggered by an RFID-activate signal from the RFID reader). While the RFID tag **120** is generally described herein as being a passive RFID tag, in other examples the RFID tag **120** may be an active RFID tag.

The utility controller **102** also includes a visual indicator **124**, which may include one or more light-emitting diodes (LEDs) or other similar illumination sources. The visual indicator **124** is viewable from outside a housing of the utility controller **102**, and the visual indicator **124** may indicate various states of the utility controller or the corresponding utilities. For example, when a utility is in the active

state (e.g., the corresponding valve is open), the visual indicator **124** may be illuminated with a first color, such as green, to indicate the utility is in the active state. When the utility is in the inactive state (e.g., the corresponding valve is closed), the visual indicator **124** may be illuminated with a second color, such as red, to indicate the utility is in the inactive state. In other examples, the color corresponding to the active state may be based on the type of utility that is being controlled. For instance, the active state of a water utility may be indicated by a blue color, and the active state of a natural gas utility may be indicated by a yellow color. By having the visual indicator **124** illuminated in both the inactive and active states of the utility, rather than simply turning off in during the inactive state, the operator is able to determine that utility controller **102** is still operational and receiving power.

In some examples, the visual indicator **124** may pulse or fade in and out while illuminated to provide additional visual effects through a sequence of changes to the output of the visual indicator **124**. When the utility controller **102** is in an emergency state or a panic state based on receiving an emergency or panic input signal, the visual indicator **124** may continuously flash the second color (e.g., red) or another color to indicate the emergency or panic situation.

The visual indicator **124** may also provide output responses from the utility controller **102** based on events. For example, if an RFID tag is detected that is unregistered, the visual indicator **124** may provide a rejection sequence, which may be a series of flashes or color sequences. As another example, when a master RFID tag is detected, the visual indicator **124** may be illuminated with a color, other than the color used for the active state and inactive state, to indicate that the utility controller **120** is in a configuration state. The indication of the configuration state may also be presented through a configuration illumination sequence (e.g., particular flashes or changes in light) that indicates the utility controller **102** is in the configuration state. A confirmation sequence may also be illuminated when a configuration RFID card is detected and the utility controller **102** setting(s) are changed. A different or similar confirmation sequence may also be illuminated when an unregistered RFID card is added as a registered card during the configuration state.

In addition or alternatively to the visual indicator **124**, the utility controller **124** may also include an audio output component (not depicted), such as a speaker, that can be controlled to emit sounds from the utility controller **124**. For instance, a first sound may be emitted when a utility is activated and a second sound may be emitted when a utility is deactivated. Similar to the visual sequences discussed above, additional or other sound sequences may also be emitted to convey additional response information, such as an unregistered card, entering a configuration mode, etc. The audio output may also sound alarms during an emergency situation. Sounds indicating startup of the utility controller may also be emitted.

The utility controller **102** also includes power conversion circuitry **116** that converts or modifies an input power **117** to the utility controller **102**. For example, the input power **117** may be 120 VAC and the power conversion circuitry **116** may convert that power to other voltages and/or currents that are suitable for the components of the utility controller **102** and/or for providing the activation signals from the output circuitry **126**. The power conversion circuitry **116** may include a transformer, which may be a step-down transformer to reduce the voltage from that of the input power **117**. The power conversion circuitry **116** may also include

AC-DC converters, DC-DC converters, voltage dividers, or the like to convert the input power **117** to power suitable for use by the components of the utility controller **102** and/or to activate the one or more valve control devices.

While FIG. 1A depicts the components of the utility controller **102** being housed within a single housing, in other examples, some components may be modularly located or positioned. As an example, the RFID reader **118** may be housed or positioned separately from the visual indicator **124**. For instance, the RFID reader **118** may be provided in a first housing that is located at an easily accessible position in the room, such as near a door or a desk, and at a height that is reachable by a user. The visual indicator **124** may be provided in a separate, or second, housing that is located away from the first housing. For instance, the visual indicator **124** may be positioned higher on a wall such that the visual indicator **124** may be seen more easily by those within the room. The remaining components of the utility controller **102** may be housed in the first housing and/or the second housing. In other examples, a second visual indicator may be provided separately from the utility controller **102**, and that second visual indicator may be positioned in a manner that can be easily seen from those in the room. In the foregoing examples, wiring between the separate housings may be provided such that signals can be communicated between the components within the housings. In other examples, the communication may be achieved wirelessly.

FIG. 1B depicts additional components of the example system **100** for controlling at least one utility with a touchless utility controller **102**. In some examples, the utility controller **102** may have additional communication circuitry to allow for longer range wireless communication, such as via wireless internet (e.g., WIFI) or other short-range communication protocols (e.g., BLUETOOTH). In such examples, the utility controller **102** may communicate with a mobile device **150** through either a short-range connection or an Internet-based connection. The mobile device **150** may have an application that allows data to be exchanged between the mobile device **150** and the utility controller **102**. For example, activation and deactivation of a utility may be initiated through user input into the application on the mobile device **150**. In addition, a duration of the shutdown timer may be displayed within an interface of the application on the mobile device **150**. Further, settings of the utility controller **102**, such as the duration of the shutdown timer, may be changed based on user input in the application on the mobile device **150**. While depicted as a smartphone in FIG. 1B, the mobile device **150** may a tablet, laptop, or other type of computing device capable of establishing the corresponding wireless connection to the utility controller **102**.

The utility controller **102** may also be communicatively connected to a server **152** via the Internet or local-area network (LAN). In some examples, multiple utility controllers **102A-C** may be connected to the server **152**. The server **152** may receive data corresponding to the current states of the utility controllers **102A-C** and/or the corresponding states of the controlled utilities (e.g., whether a utility is in an active or inactive state). The state data may be transmitted from each of the utility controllers **102A-C** to the server **152** at a regular interval or in response to a request or query generated from the server **152**.

The server **152** may also store an event log from each of the utility controllers **102A-C**. For instance, each utility controller **102A-C** may separately store a log of events that occur at the utility controller, such as when utilities are activated/deactivated and the unique identifiers of the RFID tags that were used to trigger such activation/deactivation.

11

Accordingly, from the event log, a determination of which RFID tags (and their corresponding operators) activated or deactivated a utility may be made. This event log may be transmitted to the server **152** for queries or investigation. By offloading the log data to the server **152**, each of the utility controllers may need to retain less data and can have less local memory/storage.

The server **152** may be accessible by one or more remote computers **154A-C**. The remote computers **154A-C** may access the server **152** through a web-based portal (e.g., a website/webpage) and/or a dedicated local application running the respective remote computer **154**. By accessing the server **152**, a remote computer is able to obtain information about the current states of one or more of the utility controllers **102A-C** and/or the event log data for one or more of the utility controllers **102A-C**. Such information may also be used to determine a duration for which a utility has been or was in an active state.

In some examples, the duration information may be used to assess charges for the use of the utility. For example, the utility controller **102** may control a gas valve that controls the flow of gas to a firepit in a common area of a residential community. Residents of the residential community may each be issued a registered RFID tag with a unique identifier. Accordingly, based on the RFID tag used to activate the valve and the firepit, a determination may be made as to which resident performed the activation. The duration of the activation may then be used to bill that resident for the corresponding use and duration of the firepit and gas.

The list of registered unique identifiers, stored on the utility controller **102**, for the RFID tags may also be managed via the mobile device and/or the remote computers via the server **152**. For instance, an operator may add or remove unique identifiers to or from the list. As another example, the server **152** may regularly push or update the list of the registered unique identifiers to the utility controller **102**. In other examples, the list of registered unique identifiers may be stored on the server **152** and/or a remote computer **154**. In such examples, when an RFID tag is scanned by the utility controller **102**, the utility controller **102** sends the corresponding unique identifier to the server **152**, where the server **152** performs the comparison to the list of registered unique identifiers, and then the server **152** sends an approve or reject indication back to the utility controller **102**. Upon receiving an approve indication, the utility controller **102** generates the activation signal to activate the utility (or ceases the activation signal to deactivate the utility).

By utilizing a server **152** or computing device **154** to help maintain or update the list of registered unique identifiers, the list may be changed more easily and multiple unique identifiers may be added or removed more quickly. For example, where the utility controller **102** is placed within a residential community and control of the corresponding utility by the residents is desired, the list of unique identifiers for RFID tags already distributed to residents (such as for opening doors). Other types of businesses or locations (e.g., schools, restaurants, laboratories offices) may also implement similar synchronizations between RFID tags used for other access-control purposes and the utility controller **102**. In such examples, the role of the person to which the corresponding RFID tags may also be taken into account and updates to the list of registered unique identifiers may be similarly implemented. For example, if a resident moves out of a residential complex or an employee leaves a company, the unique identifiers corresponding to the RFID tags issued to that person may be removed from the list of registered unique identifiers. As another example, if a teacher changes

12

roles and is no longer working in laboratory (e.g., changes from science teacher to history teacher), the unique identifier for that teacher's RFID tag may be removed from the list of registered unique identifiers. Such removals (and/or additions) may be performed automatically. For instance, when a new RFID tag is issued to a person having a specific role, the list of registered unique identifiers may automatically be updated to add the unique identifier of the new RFID tag. Similarly, when a person is removed from a role, the list of registered unique identifiers may be automatically be updated to remove of a unique identifier of the RFID tag issued to that person.

FIG. 2 depicts a front view of a touchless utility controller **202** with a visual indicator **206** in an inactive (or "off") state. As discussed above, the utility controller **202** is streamlined to provide a touchless form that still functions and clearly conveys state information to the operator. To achieve those benefits, a front face **204** of the utility controller **202** does not include any physically activated interface elements such as elements that are activated by physical touch (e.g., capacitive or resistive touch sensors) or physical interaction from an operator (e.g., buttons, switches, key receptacles). The term "touchless," however, does not preclude a user from touching the RFID tag against a surface of the controller **202** while bringing the RFID tag near the utility controller. Rather, the front face **204** includes a large visual indicator **206** that indicates the state of the utility controller **202** and/or the controlled utility, and an RFID reader behind the surface of the front face **204** allows for selective activation/deactivation of the utilities. The visual indicator **206** may be formed in part by a translucent portion in the front face **204** that allows for an illumination source of the visual indicator **206** to transmit light through, or illuminate, the translucent portion. Of note, while the term translucent is used herein, the translucent portion may also be transparent. The portion of the front face **204** that is illuminated by the illumination source of the visual indicator may be referred to as the illuminated portion.

The illuminated portion may occupy a substantial portion of the surface area of the front face **204**. For example, the surface area of the illuminated portion may occupy at least 10%, 20%, or 30% of the surface area of the front face **204**. By having the illuminated portion occupy a larger surface area, the ability to see the visual indicator **206** from a further distance is increased, which allows an operator to see the visual indicator **206** and determine the corresponding state of the utility from across the room.

The illuminated portion may also be near the perimeter of the front face **204** to allow for the RFID reader to be placed more toward the center of the utility controller **202** without interference from other electronics, such as the visual indicator **206**. For instance, the front face **204** may have a width W (or diameter in the case of a circularly shaped front face **204**). The outer edges, or perimeter, of the illuminated portion may be within 10%, 20%, or 30% of the width W (or diameter) from all exterior edges of the front face **204**. For instance, a distance (D_{OE}) from the exterior edge of the front face **204** to the outer edge of the illuminated portion may be less than 10%, 20%, or 30% of the width W (or diameter). In some examples, the inner edges, or perimeter, of the illuminated portion may be within 20%, 30%, 40%, or 50% of the width (or diameter) from all exterior edges of the front face **204**. For instance, a distance (D_{IE}) from the exterior edge of the front face to the inner edge of the illuminated portion may be less than 20%, 30%, 40%, or 50% of the width W (or diameter). The distances D_{OE} and D_{IE} may be the shortest distance between the two respective points and/or

13

may be measured in a direction orthogonal from an exterior edge of the front face **204**. With such a configuration, the RFID reader may be placed near the center of the front face **204** and inside (e.g., more towards the center) of the inner boundary of the illuminated portion formed by the inner edge of the continuous illuminated portion. In some examples, the shape of the illuminated portion is substantially the same as the outer boundary of the front face **204**. For instance, in the example depicted in FIG. 2, the illuminated portion forms a rectangle as does the outer boundary of the front face **204**.

While the front face **204** in this example is depicted as planar and rectangular, the front face **204** may be formed in other shapes. For instance, as mentioned above, the front face **204** may have a circular or oval perimeter. The front face **204** may also bulge outward or protrude inward to form a saucer shape, a dome shape, or a disk shape, among other types of shapes. Accordingly, the term “face” is not intended to limit the physical structure to a particular shape. In addition, while the visual indicator **206** is shown as being on the front face **204**, the visual indicator **206** may be provided on other positions of the utility controller **202**. In an example, the visual indicator **206** may project light outwards from the utility controller **202**, such as from a perimeter or back of the controller, to illuminate a portion of the wall to which the controller **202** is mounted or some other external object.

The visual indicator **206** may illuminate to convey different states or events of the utility controller **202** and/or controlled utility, as discussed herein. In FIG. 2, the visual indicator **206** is illuminated in a first color (e.g., red) to indicate that a utility being controlled is in an inactive (or “off”) state. FIG. 3 depicts a front view of the touchless utility controller **202** with the visual indicator **206** in an active (or “on”) state. For instance, the visual indicator **206** is illuminated with a second color (e.g., green) to indicate that the utility being controlled is in an active state.

FIG. 4 depicts a top view of the utility controller **202** showing a top surface **208** of the utility controller **202**. FIG. 5 depicts a left side view of the utility controller showing a left surface **210** of the utility controller **202**. FIG. 6 depicts a right side view of the utility controller **202** showing a right surface **212** of the utility controller. FIG. 7 depicts a bottom view of the utility controller **202** showing a bottom surface **214** of the utility controller **202**. The front face **204**, top surface **208**, left surface **210**, right surface **212**, the bottom surface **214**, and a back surface (not shown) form a housing that may house or encompass the components of the utility controller shown in FIG. 1A and discussed above. For instance, the components of the utility controller **202** may be mounted to a circuit board that is mounted inside the housing. The housing of the utility controller **202** may be in the form of a square or rectangular box, as depicted in FIGS. 2-7. In other examples, the front face **204** may be round and the sides of the housing are formed by a hoop or cylindrical surface. Other shapes are also contemplated, including rounded, saucer-shaped, dome-shaped, disk-shaped, or other types of shapes that can house the components discussed herein. The top surface **208**, the left surface **210**, the right surface **212**, and the bottom surface **214** may collectively form or be referred to as an outer side perimeter of the housing.

As can be seen in FIGS. 4-7, the outer side perimeter of the housing of the utility controller **202** also may not include any buttons, key receptacles, or other physically activated interface elements. The back surface may also not include any such elements.

14

The bottom surface **214** may include one or more through holes to allow for wiring, such as input power wiring and output transmission wires, to go into and/or out of the housing. In the example depicted, the bottom surface **214** may form a first through hole **216** may be for the power input transmission wires (e.g., power cord) and a second through hole **218** for the output transmission wires (e.g., wires from the output circuitry to the valve control devices). In other examples, a single through hole may be used for both types of wires to reduce the number of through holes in the housing. While the through holes are depicted as being in the bottom surface **214**, in other examples, the through hole(s) may be formed in any portion of the outer side perimeter or on the back surface of the housing of the utility controller **202**.

FIG. 8 depicts a front view of the touchless utility controller **202** with a visual indicator **206** that changes with a shutdown timer. As discussed herein, when the utility is activated from a detection of a registered RFID tag, a shutdown timer may be initiated. As the shutdown timer decrease as time passes, it is useful to convey this information to the operator so that the operator is aware how much time is left on the shutdown timer. FIG. 8 shows the visual indicator **206** in the active state (e.g., illuminated with green) when the shutdown timer is first started and no time has passed. FIG. 9 depicts a front view of the utility controller **202** and the visual indicator **206** after a partial duration of the shutdown timer has passed. As can be seen in FIG. 9, a segment **210** of the visual indicator **206** (e.g., the illuminated portion) is no longer illuminated. The length or size of the non-illuminated segment **210** may correspond to the amount of time that has passed since the shutdown timer was initiated.

If the operator sees that the shutdown timer is about to expire and desires more time, in some examples, the operator may be able to rescan to the registered RFID tag to reset the shutdown timer to its full duration. For instance, after a partial duration of the shutdown timer has passed, the utility controller **202** may subsequently detect a double scan of the registered RFID tag to reset the shutdown timer to its full duration. The double scan may be a detection of the same registered RFID card within a time threshold (e.g., 1-5 seconds). When the double scan is detected, the shutdown timer may be reset instead of deactivating the utility and then reactivating the utility. Other types of reactivation scan sequences of the RFID tag may be possible as well to reset the shutdown timer to its full duration.

The utility controller **202** may support a plurality of control mechanisms to open the valve, reset the shutdown timer, and close the valve, all with a single registered RFID tag or with multiple RFID tags. For example, while the utility valve is closed, utility controller **202** may provide an activation signal (e.g., ON) responsive to a scan of the registered RFID tag, causing a utility valve to open. While the utility valve is open, the utility controller **202** may reset the shutdown timer responsive to a second and/or later scan of the same registered RFID or another registered RFID tag (e.g., RESET TIME). In another or additional example, while the utility valve is open, the utility controller **202** may reset the shutdown timer responsive to an operator holding the same registered RFID tag or another registered RFID tag (e.g., RESET TIME) near to utility controller **202** for at least a threshold time period (e.g., 1-5 seconds). In another or additional example, while the utility valve is open, the utility controller **202** may cease transmission of the activation signal responsive to an operator holding the same registered RFID tag or another registered RFID tag (e.g., OFF) near to

15

utility controller **202** for at least a threshold period of time period (e.g., 1-5 seconds). In some examples, detecting the registered RFID tag near the utility controller **202** for the threshold time period may be interpreted as equivalent to detecting a double-scan of the registered RFID tag. For example, if the utility controller detects two scans of the registered RFID tag within a threshold period of time (or a continuous scan of the registered RFID tag for a threshold period of time), the utility controller **202** may (depending on implementation) either reset the shutdown timer or cease transmission of the activation signal. In examples, the threshold period for detecting a continuous scan and the threshold period for detecting a double-scan may be the same or they may be different. Other control mechanisms utilizing one or multiple RFID tags to provide the activation signal, reset the shutdown timer, and cease providing the activation signal are possible and contemplated.

FIGS. **10-11** depict front views of the touchless utility controller **202** with a visual indicator **206** and a shutdown timer indicator **220** on the front face **204**. In some examples, the visual indicator **206** that indicates the state of the utility may not indicate or convey information about the time remaining on the shutdown timer. Instead, a separate shutdown timer indicator **220** may be incorporated on the front face **204** of the utility controller **202** to indicate the amount of time remaining on the shutdown timer. In the example depicted in FIGS. **10-11**, the shutdown timer indicator **220** is an illuminated ring that may be formed or manufactured similarly to the visual indicator **206**. FIG. **10** shows the shutdown timer indicator **220** when the shutdown timer is first started and no time has passed. FIG. **11** shows the shutdown timer indicator **212** after a partial duration of the shutdown timer has passed. As can be seen in FIG. **11**, a segment **213** of the shutdown timer indicator **212** is no longer illuminated. The length or size of the non-illuminated segment **213** corresponds to the amount of time that has passed since the shutdown timer was initiated.

In other examples, the shutdown timer indicator **220** may have different forms or have a different geometric shape where the illuminated portion of the geometric shape changes (e.g., reduces) as time passes and the time remaining on the shutdown timer decreases. In yet other examples, the shutdown timer indicator **220** may be a numerical indicator (e.g., seven-segment display or the like) that indicates an amount of time remaining on the shutdown timer in a numerical format.

For example, the shutdown timer indicator **220** may display a numerical indicator (e.g., seven-segment display) to display the amount of time that the shutdown timer was set for upon detection of a registered RFID tag. For example, if a "four hour" registered RFID tag is detected, the shutdown timer indicator may initially display "4:00" and continue to display the numerical indicator of the time remaining as it counts down for an initial display period (e.g., five seconds), after which, in examples, the numerical indicator may no longer be displayed. In other examples, the numerical indicator may continue to be displayed for as long as there is time left on the countdown timer. In some examples where the numerical indicator disappears after the initial period, the numerical indicator may reappear at certain points during a countdown. For example, the shutdown timer indicator **220** may display the numerical indicator of the remaining time when the countdown reaches certain preset limits (e.g., one hour left, ten minutes left, one minute left, etc.). Again, the numerical display of the remaining time may remain displayed for a preset period before disappearing. In other examples, the numerical display may

16

remain displayed until the countdown reaches zero (e.g., it may reappear with one minute (or some other time period) left and remain until the countdown timer reaches zero) or until a registered RFID tag is detected to reset the time period. In some examples, the numerical indicator is displayed simultaneously with a graphical shutdown timer indicator, such as the illuminated ring shown in FIGS. **10** and **11**. In other examples, only one of the numerical indicator or the graphical shutdown timer indicator is displayed at a time.

As discussed, mobile device **150** (FIG. **1B**) may have an application that allows data to be exchanged between the mobile device **150** and the utility controller **102**. A duration of the shutdown timer may be displayed within an interface of the application on the mobile device **150** in a similar manner as discussed with respect to FIGS. **8, 9, 10**, and **11**. For example, a shutdown timer within an application on mobile device **150** may cause the display on the mobile device **150** of a similar graphical and/or numerical indicator as shutdown timer indicator **220**. In examples where the mobile device **150** comprises a registered RFID tag that is detected by touchless utility controller **102** to initiate a shutdown timer, the initial shutdown timer duration (and/or remaining shutdown timer duration) may be displayed on the mobile device in a similar manner as described with respect to shutdown timer indicator **220**.

FIG. **12** depicts a front view of a touchless utility controller **1202** with multiple RFID readers and visual indicators for different utilities. In some examples, the utility controller **1202** is capable of selectively controlling different utilities through different activation points on the front face **1204** of the utility controller. For instance, a first portion of the front face **1204** may include a first visual indicator **1206** and a first RFID reader **1208**, a second portion of the front face **1204** may include a second visual indicator **1210** and a second RFID reader **1212**, a third portion of the front face **1204** may include a third visual indicator **1214** and a third RFID reader **1216**, and a fourth portion of the front face **1204** may include a fourth visual indicator **1218** and a fourth RFID reader **1220**. While multiple RFID readers are shown as being placed in the various positions, the functionality may be achieved by including different antennas for an RFID reader at each corresponding location and using common RFID reader circuitry. Also, it should be appreciated that the respective RFID readers/antennas are located within the housing or on the interior of the front face **1204** and the positions shown in FIG. **12** are for reference as to relative position to the front face **1204**.

Each of the RFID readers and visual indicators in FIG. **12** may function substantially similarly to the RFID readers and visual indicators discussed above but for control of separate valves and/or utilities. For instance, scanning a registered RFID tag at the position of the first RFID reader **1208** causes the utility controller **1202** to send an activation signal to a first valve to control a first utility. When the corresponding first utility is in the active state, the first visual indicator **1206** is illuminated with a first color (e.g., green) indicating the active state of the first utility/valve. When the corresponding first utility is in the inactive state, the first visual indicator **1206** is illuminated with a second color (e.g., red) indicating the inactive state of the first utility/valve. Similarly, scanning a registered RFID tag at the position of the second RFID reader **1212** causes the utility controller **1202** to send an activation signal to a second valve to control a second utility, which may be the same or different type of utility as the first

utility. The second visual indicator **1210** is similarly illuminated to convey state information about the second utility/valve.

In some examples, a fifth or full-control RFID reader **1220** may be included that may be used for activation or deactivation of all the utilities within the control of the utility controller **1202**. For example, scanning a registered RFID card at the position of the full-control RFID reader **1220** may result in activation or deactivation of all the controlled utilities/valves. Upon such activation/deactivation, all the visual indicators **1206**, **1210**, **1214**, **1214** may be illuminated with the corresponding changed state of the utility.

The different utilities corresponding to the different RFID readers may be controlled with common RFID tags or separate RFID tags. In an example where a common RFID tag may be used, one RFID tag may be used to activate/deactivate any of the utilities controlled by the utility controller **1202**. In other examples, one RFID tag may be able to control only one (or a subset) of the available utilities and another RFID tag may be able to control another (or another subset) of the available utilities. For example, a scan of an RFID tag at the first RFID reader **1208** may cause activation/deactivation of the first utility, but a scan of that same RFID tag at the second RFID reader **1212** may not cause activation/deactivation of the second utility. Instead, a second RFID tag may be needed to activate/deactivate the second utility.

Where different utilities are controlled by the different RFID readers, the colors displayed by the respective visual indicators may be different to indicate the different type of utility being controlled. As an example, the first visual indicator **1206** may display a blue color when active to indicate that a water utility is being controlled through interaction with the first RFID reader, and the second visual indicator **1210** may display a yellow color when active to indicate that a gas utility is being controlled.

FIG. 13 depicts an example method **1300** for controlling a utility with a touchless utility controller. At operation **1302**, an RFID signal is received or detected by an RFID reader of the utility controller. The RFID signal is received from an RFID tag of a set of RFID tags. The RFID signal includes a unique identifier for the RFID tag from which the RFID signal was received.

At operation **1304**, the received RFID signal is compared to a list or database of registered unique identifiers. The database or list of unique identifiers may be stored in the local memory of the utility controller or remotely from the utility controller. At operation **1306**, based on the comparison performed in operation **1304**, a determination is made as to whether the received RFID signal corresponds to a registered RFID tag. For example, if the unique identifier of the received RFID signal matches a unique identifier in the list of registered unique identifiers, the RFID tag may be determined to be a registered RFID tag. If the unique identifier of the received RFID signal does not match a unique identifier in the list of registered unique identifiers, the RFID tag may be determined to not be a registered RFID tag.

If the RFID tag is determined to not be a registered RFID tag in operation **1306**, the method **1300** flows to operation **1308** where a rejection sequence is performed. The rejection sequence may be a flashing of the visual indicator or other illumination sequence of the visual indicator to indicate that the detected RFID tag has been rejected. An audible indicator may also or alternatively be sounded. If the RFID tag is determined to be a registered RFID tag in operation **1306**, the method **1300** flows to operation **1310**.

At operation **1310**, in response to the determination that the RFID tag is a registered RFID tag, an activation signal is transmitted from the utility controller to a valve control device of a valve that controls a utility (e.g., the flow of water or gas). Transmission of the activation signal causes the corresponding valve to open to allow the flow of the fluid, such as gas or water.

At operation **1312**, a shutdown timer is also initiated. For instance, upon activating the utility, the shutdown timer begins for a set duration. The set duration of the shutdown timer may be a setting of the utility controller. In other examples, the set duration of the shutdown timer may be dependent on the RFID card that was scanned. For example, one RFID tag may cause activation of a utility and a shutdown timer of a first set duration to initiate, and another RFID tag may cause activation of a utility and a shutdown timer of a second set duration to initiate. The value for the set duration may be included in the data of the RFID signal received from the RFID tag, or the value for the set duration may be accessed via a query or lookup function. For instance, a first set of RFID tags may be associated in a list or database with a first set duration, a second set of RFID tags may be associated with a second set duration, and a third set of RFID tags may be associated with a third set duration.

In addition, at operation **1314**, the visual indicator is illuminated to an active or "on" indication to indicate the active state of the utility. For instance, the visual indicator may be illuminated with a first color (e.g., green) to indicate the corresponding utility is active.

At operation **1316**, upon expiration of the shutdown timer, the utility controller causes the corresponding utility to become inactive (e.g., cause the valve to close). For instance, the transmission and/or generation of the activation signal may be ceased. Cessation of the activation signal causes the utility to be inactive. For instance, cessation of the activation signal causes the valve to close, which prevents the flow of fluid through the valve. In some examples, the activation signal is transmitted for the entire duration of the shutdown timer (unless another deactivation condition occurs prior to the expiration of the shutdown timer). At operation **1318**, upon inactivation of the corresponding utility, the visual indicator is illuminated to an inactive or "off" indication to indicate the inactive state of the utility. For instance, the visual indicator may be illuminated with a second color (e.g., red).

FIG. 14 depicts an example method **1400** for configuring a touchless utility controller. At operation **1402**, an RFID signal is received or detected by an RFID reader of the utility controller. The RFID signal is received from an RFID tag of a set of RFID tags. The RFID signal includes a unique identifier for the RFID tag from which the RFID signal was received.

At operation **1404**, the received RFID signal is compared to a list or database of registered unique identifiers for master RFID tags. The database or list of unique identifiers of master RFID tags may be stored in the local memory of the utility controller or remotely from the utility controller. At operation **1406**, based on the comparison performed in operation **1404**, a determination is made as to whether the received RFID signal corresponds to a master RFID card. For example, if the unique identifier of the received RFID signal matches a unique identifier in the list of unique identifiers for master RFID tags, the RFID tag may be determined to be a master RFID tag. If the unique identifier of the received RFID signal does not match a unique

identifier in the list of unique identifiers for master RFID tags, the RFID tag may be determined to not be a master RFID tag.

If the RFID tag is determined to not be a master RFID tag in operation 1406, the method 1400 may end or may flow to operation 1313 of method 1300 in FIG. 13 where the received RFID signal is compared to a list of unique identifiers for activation/deactivation RFID tags. If the RFID tag is determined or detected to be a master RFID tag in operation 1406, the method 1400 flows to operation 1408.

At operation 1408, in response to the determination that the RFID tag is a master RFID tag, the utility controller enters a configuration mode. Entering the configuration mode may also include illuminating a configuration sequence or a third color on the visual indicator to indicate that the utility controller has entered the configuration mode. During the configuration mode, the utility controller awaits a scan or detection of a configuration RFID card. The configuration mode may have a set duration (e.g., 10 seconds), and upon expiration of the set duration, the utility controller exits the configuration mode and returns to its prior state before the master RFID tag was detected.

At operation 1410, a configuration signal RFID signal is received by the RFID reader from a configuration RFID tag (e.g., a configuration RFID tag is detected). The configuration RFID signal may include a unique identifier and/or configuration data for one or more settings. For example, based on the unique identifier in the configuration RFID signal from the configuration RFID tag, the utility controller 102 may perform a local lookup to determine a corresponding settings change that should be made. In other examples, the configuration RFID signal may include data directly indicating the settings change that should be made. For instance, the configuration RFID data may include a set duration (e.g., 4 hours, 6 hours, 8 hours, 12 hours) for the shutdown timer. Accordingly, scanning different configuration RFID cards may cause different settings to be changed or the same setting to be changed to a different value. At operation 1412, the utility controller reconfigures the settings based on the configuration RFID signal that was received in operation 1410.

In some examples, while in the configuration mode, new RFID tags may be registered with utility controller 102. For example, while the utility controller is in the configuration mode, a new RFID tag with a new unique identifier may be scanned (e.g., detected) and the that new unique identifier may be added to the list of registered unique identifiers. Thus, after the registration occurs, the new RFID tag may be used to activate or deactivate the utility controlled by the utility controller.

While RFID technology is primarily discussed as being used above, in some examples other forms of wireless communication may be used. For instance, longer range radio-frequency communication may also be used that operate in the ultra-high frequency range (UHF) (e.g., 300 MHz to 3 GHz). The radio frequency may operate in the band between 300 MHz to 1 GHz or 300 MHz to 400 MHz. In such examples, instead of RFID tags being used for activation, RF remote controls may be used, with each RF remote control emitting a unique identifier to allow for distinguishing remote in a manner similar to distinguishing RFID tags. The utility controller may then include a corresponding RF receiver. In other examples, the wireless communication may be accomplished by infrared control. For instance, instead of RFID tags being used for activation, infrared remote controls may be used, with each infrared remote control emitting a unique identifier to allow for distinguish-

ing the remote controls in a manner similar to distinguishing RFID tags. The utility controller may then include a corresponding infrared receiver. More generally, the utility control systems may include a wireless receiver (e.g., RFID reader, RF receiver, IR receiver) and a set of wireless transmitters (e.g., RFID tags, RF remotes, IR remotes).

The embodiments described herein may be employed using software, hardware, or a combination of software and hardware to implement and perform the systems and methods disclosed herein. Although specific devices have been recited throughout the disclosure as performing specific functions, one of skill in the art will appreciate that these devices are provided for illustrative purposes, and other devices may be employed to perform the functionality disclosed herein without departing from the scope of the disclosure. In addition, some aspects of the present disclosure are described above with reference to block diagrams and/or operational illustrations of systems and methods according to aspects of this disclosure. The functions, operations, and/or acts noted in the blocks may occur out of the order that is shown in any respective flowchart. For example, two blocks shown in succession may in fact be executed or performed substantially concurrently or in reverse order, depending on the functionality and implementation involved.

This disclosure describes some embodiments of the present technology with reference to the accompanying drawings, in which only some of the possible embodiments were shown. Other aspects may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments were provided so that this disclosure was thorough and complete and fully conveyed the scope of the possible embodiments to those skilled in the art. Further, as used herein and in the claims, the phrase “at least one of element A, element B, or element C” is intended to convey any of: element A, element B, element C, elements A and B, elements A and C, elements B and C, and elements A, B, and C. Further, one having skill in the art will understand the degree to which terms such as “about” or “substantially” convey in light of the measurement techniques utilized herein. To the extent such terms may not be clearly defined or understood by one having skill in the art, the term “about” shall mean plus or minus ten percent.

Although specific embodiments are described herein, the scope of the technology is not limited to those specific embodiments. Moreover, while different examples and embodiments may be described separately, such embodiments and examples may be combined with one another in implementing the technology described herein. One skilled in the art will recognize other embodiments or improvements that are within the scope and spirit of the present technology. Therefore, the specific structure, acts, or media are disclosed only as illustrative embodiments. The scope of the technology is defined by the following claims and any equivalents therein.

What is claimed is:

1. A system for controlling a utility, the system including:
 - an RFID reader;
 - a processor, and memory storing a list of registered unique identifiers and storing instructions that, when executed the processor, cause the system to perform operations including:
 - receiving, by the RFID reader, a first RFID signal from an RFID tag, the first RFID signal including a unique

21

identifier, wherein the RFID tag is one of a set of
 RFID tags that includes a master RFID tag and a
 configuration RFID tag;
 comparing the unique identifier to the list of registered
 unique identifiers to determine that the RFID tag is
 a registered RFID tag;
 in response to determining the RFID tag is a registered
 RFID tag:
 transmitting an activation signal to a valve control
 device to cause a valve controlling a utility to
 open;
 receiving, by the RFID reader, a second RFID signal
 from the master RFID tag;
 entering a configuration mode based at least in part on
 receiving the second RFID signal;
 while in the configuration mode, receiving, by the
 RFID reader, a third RFID signal from the configu-
 ration RFID tag; and
 based at least in part on receiving the third RFID signal,
 modifying one or more settings of the system.

2. The system of claim 1, wherein the master RFID tag
 includes a unique identifier that, when received by the RFID
 reader, causes the system to enter the configuration mode,
 wherein the system remains in the configuration mode for a
 set duration or until an additional RFID tag is scanned.

22

3. The system of claim 1, wherein the configuration mode
 allows for the addition of unregistered RFID tags to the list
 of registered unique identifiers, wherein each unregistered
 RFID tag scanned during the configuration mode is stored in
 the memory as a registered RFID tag.

4. The system of claim 1, wherein the configuration RFID
 tag includes configuration data that, when received by the
 RFID reader, causes the processor to modify one or more
 settings of the utility controller based on the configuration
 data.

5. The system of claim 4, wherein the configuration RFID
 tag is used to change a shutdown timer duration of the utility
 controller, and the configuration data includes a specific
 duration for the shutdown timer.

6. The system of claim 1, wherein the master RFID tag,
 when scanned for a duration exceeding a reset threshold,
 causes the processor to perform a factory reset of the utility
 controller.

7. The system of claim 6, wherein the factory reset
 includes clearing the memory of registered identifiers for
 RFID tags, excluding the master RFID tag and a predeter-
 mined subset of RFID tags.

8. The system of claim 1, further comprising:
 illuminating a visual indicator to a first color when the
 master RFID tag is scanned.

* * * * *